TRACTIAN
INDUSTRIAL INTELLIGENCE

TMP64

Industrial I²C Digital Temperature Sensor

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GENERAL DESCRIPTION

The **TMP64** is a 32-bit resolution digital temperature sensor designed for industrial process monitoring applications. It communicates over a standard I²C bus at the fixed address **0x48** and provides temperature data as a 32-bit signed integer in milli-Celsius units.

The device features a fixed identification register (`WHO_AM_I`) at address `0x0F` that can be queried at startup to confirm sensor presence. Temperature data is accessible via a 4-byte sequential read starting at register address `0x00`.

FEATURES & BENEFITS

- I²C interface — fixed address **0x48**
- 32-bit signed temperature output in **milli-Celsius**
- Measurement range: −40 °C to +125 °C
- Accuracy: ±0.5 °C (−10 °C to +85 °C)
- WHO_AM_I = `0xA5` at register address `0x0F`
- 4-byte sequential read from register `0x00`, MSB first
- Open-drain SDA/SCL lines
- 1 Hz measurement update rate (nominal)
- Operating voltage: 3.0 V – 3.6 V
- Low standby current: < 5 µA

PIN DESCRIPTION

PIN	NAME	TYPE	DESCRIPTION
1	GND	PWR	Ground reference
2	VDD	PWR	Supply voltage, 3.0–3.6 V
3	**SDA**	**I/O OD**	**I²C serial data line — open-drain**
4	**SCL**	**I OD**	**I²C serial clock line — open-drain**
5	ALERT	O OD	Overtemperature alert output — open-drain

DC ELECTRICAL CHARACTERISTICS

T_A = 25 °C, V_{DD} = 3.3 V, unless otherwise noted.

PARAMETER	SYMBOL	MIN	TYP	MAX	UNIT
Supply Voltage	V_{DD}	3.0	3.3	3.6	V
Active Supply Current	I_{DD}	—	210	350	µA
Standby Current	I_{STBY}	—	2	5	µA
SDA/SCL Logic-Low Input	V_{IL}	−0.3	—	0.8	V
SDA/SCL Logic-High Input	V_{IH}	2.1	—	V_{DD}+0.3	V
SDA Output Low Voltage	V_{OL}	—	—	0.4	V
Measurement Range	—	−40	—	+125	°C
Temperature Accuracy	**ε_T**	**—**	**±0.2**	**±0.5**	**°C**

PARAMETER	SYMBOL	MIN	TYP	MAX	UNIT
I²C Clock Frequency	f_{SCL}	—	—	400	kHz
Measurement Update Rate	f_{CONV}	—	1	—	Hz

I²C TIMING

Figure 1 — Write-then-Read Sequence (Register Read)

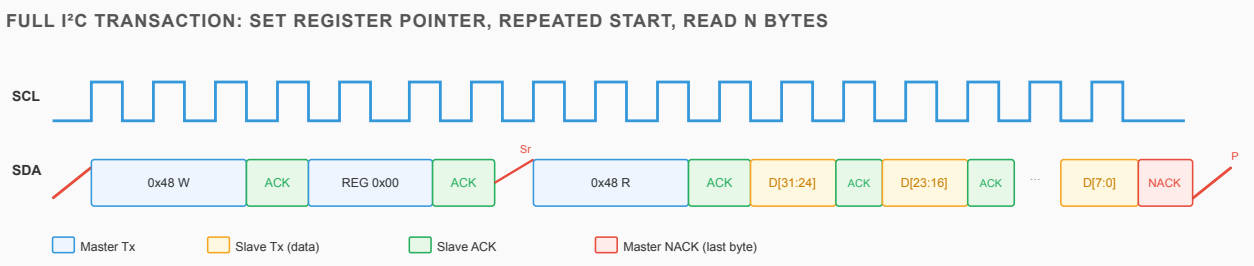

Figure 2 — Protocol Symbol Reference

SYMBOL	MEANING	DRIVEN BY
`S`	START — SDA falls while SCL high	Master
`Sr`	Repeated START — no STOP between write and read phases	Master
`P`	STOP — SDA rises while SCL high	Master
`A`	ACK — slave pulls SDA low during 9th clock	Slave
`N`	NACK — master releases SDA high on last byte	Master
`ADDR W/R`	7-bit address + direction bit (0 = write, 1 = read)	Master
`D0…Dn`	Data bytes, MSB first within each byte	Slave

REGISTER MAP

ADDRESS	NAME	ACCESS	DESCRIPTION
`0x0F`	**WHO_AM_I**	**RO (1 byte)**	**Fixed device identification byte. Always reads `0xA5`.**
`0x00`	**TEMP_MSB**	**RO**	**Temperature bits [31:24]. Sequential read from `0x00` returns all 4 bytes in order.**
`0x01`	**TEMP_B2**	**RO**	**Temperature bits [23:16].**
`0x02`	**TEMP_B1**	**RO**	**Temperature bits [15:8].**
`0x03`	**TEMP_LSB**	**RO**	**Temperature bits [7:0] — least significant byte.**
`0x04-0x0E`	RSVD	—	Reserved. Do not access.
`0x10-0xFF`	RSVD	—	Reserved. Do not access.

0x0F — WHO_AM_I Device Identification

RO · Fixed: 0xA5

7	6	5	4	3	2	1	0
1	**0**	**1**	**0**	**0**	**1**	**0**	**1**

Fixed value `0xA5` (binary: 1010 0101). Used to confirm device presence and successful I²C communication on startup.

0x00–0x03 — TEMP[31:0] Temperature Data (32-bit, MSB first)

RO · 4 bytes sequential

byte 0	byte 1	byte 2	byte 3				
T[31:24]	**T[23:16]**	**T[15:8]**	**T[7:0]**				

32-bit signed two's complement integer in **milli-Celsius**. Divide by 1000 to obtain °C. Master ACKs bytes 0–2; master NACKs byte 3 to terminate the transfer.

RAW VALUE (INT32)	TEMPERATURE	HEX (32-BIT)
−40000	−40.000 °C	`0xFFFF6360`
0	0.000 °C	`0x00000000`
25000	**25.000 °C**	**`0x000061A8`**
85000	85.000 °C	`0x00014C08`
125000	125.000 °C	`0x0001E848`

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